

The Witting Architecture

An Ultimate Substrate-Coherent Computer:
Content-Addressed Memory, Group-Action Compute,
and BIER-Multicast Fabric at 70M TPS over 10^9 Nodes

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In collaboration with the UOR Foundation, Oko, Smart Assets, and DTR/HLIX

June 2026 *v2*

Abstract

Computer architecture is stuck. The Von Neumann bottleneck dominates every modern deployment; cache hierarchies are heuristic; ISAs accrete by accident; distributed coordination is layered on after the fact; trust is retrofitted; and the binary substrate wastes 58.5% of its information density relative to the ternary minimum at every memory cell. This paper proposes a complete redesign—hardware and software—arrived at by taking three independent recent stacks seriously at once: Oko’s network-fabric Byzantine Lattice Agreement consensus (measured at 12M TPS per server), the UOR Foundation’s Atlas-12288 content-addressed substrate (48 pages $\times$ 256 bytes with R_{96} resonance classification), and the HLIX/HCP projection cycle. We show these three *converge* to a single optimal point: a 40-vertex Witting compute lattice realizing the symplectic polar space $W(3,3)$, with $\text{Aut}(W(3,3)) \cong \text{Sp}(4, \mathbb{F}_3) \cong W(E_6)$ of order 51,840 as its instruction set; ternary native arithmetic; Bose–Mesner three-channel spectral memory hierarchy; Bell-qutrit storage cells with the CSS $[[240, 81, 4, 3]]_3$ code as native ECC; and BIER multicast as the fabric replacing the bus entirely. The natural information density is $3/8 = q/2^q$, which we show is independently the R_{96} compression ratio, the CSS code rate, and the chiral-eigenspace fraction $g/v = 15/40$. The substrate’s natural cadence is $j(i) = 1728 = k^3$ per consensus epoch, giving $70\text{M TPS} \times 1728 = 1.21 \times 10^{11}$ orbit-cycles per second. A $40^6 = 4.1$ billion-node fractal deployment exceeds the planetary 10^9 -node target by a factor of four. The architecture eliminates: the bus, the von Neumann bottleneck, the ISA accretion problem, the cache hierarchy, separate networking, separate ECC, separate cryptography, separate consensus, and root-user trust assumptions. Every primitive is forced by the same substrate; nothing is retrofitted.

The Witting Architecture. *Compute is a group action on a content-addressed substrate. The group is $\text{Aut}(W(3,3)) = \text{Sp}(4, \mathbb{F}_3) = W(E_6)$. The substrate is the Atlas-12288 frame. Memory, ISA, cache, network, ECC, identity, and consensus are not separate subsystems; they are four facets of the same algebra evaluated at four scales.*

Contents

1 Introduction: The End of the Von Neumann Era

Computer architecture has not changed structurally since 1945. The Von Neumann model — separated CPU and memory, communicating over a serial bus, executing instructions one at a time

on byte-addressed memory—still underwrites every smartphone, every server, every supercomputer. The improvements since then are incremental: pipelining, caches, out-of-order execution, SIMD lanes, GPGPU acceleration, and the recent retreat into chiplet packaging when monolithic die scaling failed. The exponential gains of Moore’s law have been amortised against a stubbornly sequential model of computation.

Three independent recent stacks each chip away at one piece of the Von Neumann legacy:

- **Oko** (the network-fabric BLA consensus protocol) eliminates the leader-based quadratic bottleneck of BFT, replacing it with BIER multicast over virtual switches; it measures 12 M TPS per server at fixed 55-validator committee and projects beyond 1 billion aggregate TPS without expanding the committee. Throughput is a per-server parameter, not a network parameter.
- **The UOR Foundation** (Universal Object Reference) defines a content-addressed substrate via canonical sha256 (ψ -pipeline canonicalisation), an ontology of 474 classes and 948 properties, and an Atlas-12288 framework of 48 pages $\times$ 256 bytes per frame with an R_{96} resonance classification that compresses 256 byte values to 96 classes.
- **HLIX** (Hybrid Linked Infrastructure Xchange) operationalises the HCP projection cycle (*Projection* $\rightarrow$ *Execution* $\rightarrow$ *Emission*) over a container store, with smart-asset workloads, agentic orchestration, and a 1 billion-node planetary mesh target.

Our finding. These three stacks converge on the same architectural fixed point. The Oko committee size 55, the UOR atlas frame 12288, the UOR resonance classification 96, and the HLIX HexaScaler axes are all monomials in a single five-element substrate-primitive generating set $\{q, \mu, F_5, \Phi_3, \Phi_6\}$ over the master integer $q = 3$. The architectural floor they all approach is the **Witting polytope**—the unique 40-vertex configuration on which the symplectic polar space $W(3, 3)$ is canonically realised—with Aut group $\text{Sp}(4, \mathbb{F}_3) \cong W(E_6)$ of order 51,840.

This paper takes that convergence as a design imperative. We propose **the Witting Architecture (W-Arch)**: a complete hardware-plus-software computer architecture in which every layer (gate, register, ISA, cache, memory, bus, network, ECC, identity, OS, workload) is derived from the same Witting polytope and its automorphism group. The result eliminates the bus, the cache hierarchy as a separate concept, the ISA accretion problem, separate ECC subsystems, retrofitted networking, retrofitted security, and the root-user trust model.

2 The Seven Axioms

A1: Content-Address Everything

Every datum in the system—memory cell, register, instruction, message, process, file—has a single canonical address derived from its content by the ψ -pipeline: a deterministic canonicalisation followed by Blake3 (or sha256) hashing. No location-based addresses exist anywhere.

A2: Compute Is Group Action

The only operation available to a process is the action of an element of $\text{Aut}(W(3, 3)) = W(E_6)$ on an edge-mode configuration $\mathcal{C} \subseteq E(W(3, 3))$. There is no separate ALU, control unit, branch unit, or vector unit. Arithmetic, control flow, and parallelism are specialisations of the same group action.

A3: Spectral Cache

Cache hierarchy is the spectral decomposition of the $W(3,3)$ adjacency operator $A = kE_0 + rE_1 + sE_2$, with the three projectors E_0, E_1, E_2 of ranks $\{1, f, g\} = \{1, 24, 15\}$ acting as the three native cache levels. Cache replacement is substrate-forced, not heuristic.

A4: Bell Qutrit Storage Cell

Each storage cell holds the maximally entangled Bell qutrit $\Omega = q^{-1/2} \sum_j j_p j_f$ across past and future temporal registers. Read is the Choi–Jamiołkowski overlap; write is projection onto the future register. This is content-addressable storage at the substrate.

A5: BIER Multicast Is the Fabric

There is no bus. Every core has 12 direct neighbours (the SRG valency $k = 12$) plus 27 indirect-but-reachable neighbours (the disjoint shell $q^q = 27$). All communication is BIER multicast through embedded virtual switches; the bus bandwidth scales linearly with core count, not as a shared resource.

A6: CSS Code Is Native ECC

The CSS code $[[|E|, q^{q+1}, \mu, q]]_3 = [[240, 81, 4, 3]]_3$ is hardwired into the memory frame. Logical-to-physical ratio is fixed at $27/80 = q/2^q = 3/8$ —which is *independently* the R_{96} compression ratio and the chiral-eigenspace fraction g/v . The substrate’s natural information density is $3/8$.

A7: Identity is a Stabilizer Subgraph

A process, user, or container has no UID, no PID, and no root capability. Its identity is the subgroup $\text{Stab}_{\text{Aut}(W(3,3))}(\mathcal{C})$ of automorphisms fixing its edge-mode configuration. Provenance is the orbit history; trust is the orbit signature; consensus is orbit-equivalence under aggregate signature.

3 The Atlas-12288 Memory Frame: The Universal Unit

3.1 Why 12288

The UOR Foundation’s atlas-12288 implementation organises addressable storage into frames of 12288 bytes structured as 48 pages of 256 bytes each, with a resonance classification mapping each 256-byte alphabet onto 96 equivalence classes. The substrate readings of these three constants are exact:

$$\begin{aligned}
 48 &= q! \cdot 2^q = 6 \cdot 8 \\
 256 &= \mu^4 = 2^{\Phi_6+1} = 4^4 \\
 96 &= 2^{F_5} \cdot q = 32 \cdot 3 \\
 12288 &= q! \cdot 2^{q+\Phi_6+1} = 6 \cdot 2048
 \end{aligned}$$

where $q = 3$, $\mu = 4$, $F_5 = 5$, and $\Phi_6 = 7$ are the substrate's five generating primes. The R_{96} compression ratio is then exactly

$$\boxed{\frac{R_{96}}{2^8} = \frac{96}{256} = \frac{3}{8} = \frac{q}{2^q}}$$

which is also the rate of the CSS code $[[240, 81, 4, 3]]_3$ used as the substrate ECC (§??), and also the s -eigenspace fraction $g/v = 15/40$ of the $W(3,3)$ spectrum. *The substrate's natural information density is $3/8$, and three different arithmetic constructions land there simultaneously.*

Universal Density Theorem

The natural information density of a $\text{Sp}(4, \mathbb{F}_3)$ -equivariant content-addressed computer is $3/8 = q/2^q$. This density is realised *independently* by:

- (i) the UOR Atlas resonance compression $R_{96}/2^8$,
- (ii) the substrate CSS code rate $q^{q+1}/|E|$,
- (iii) the chiral-eigenspace fraction g/v of the $W(3,3)$ adjacency.

Any architecture deviating from $3/8$ leaves substrate-natural information unused.

3.2 Atlas frame layout

An Atlas-12288 frame is the unit of memory addressing in W-Arch. Its layout:

Field	Size / meaning
Frame header	32 bytes (sha256 self-reference + version)
Page directory	48 entries $\times$ 4 bytes = 192 bytes (page CIDs)
Pages 0–47	48×256 bytes = 12288 bytes payload
R_{96} classifier table	per-page 96-bit mask (which classes are active)
CSS parity	7 qutrit syndromes per 240-qutrit block ($\mu = 4$ distance)
Aggregate signature	MuSig 64 bytes (frame attestation)

The frame is content-addressed by Blake3 over its canonical serialisation (Cap'n Proto in the Oko stack), with frame CID a 32-byte digest. The 64-bit flat address space decomposes as $40 \cdot 1296 \cdot 2^{49.2}$, where the first factor selects one of the 40 Witting vertices (Sylow-3 subgroup of Aut, by Theorem ??), the second selects an element of its normaliser, and the residual 49.2 bits encode the frame's contingent payload.

Theorem 1 (Hidden Sylow Address Bijection). $n_3(\text{Sp}(4, \mathbb{F}_3)) = v = 40$. *The 40 Witting vertices are in canonical bijection with the 40 Sylow-3 subgroups of $\text{Aut}(W(3,3))$. Every W-Arch object address selects a unique Sylow-3 coset.*

4 Hardware: The Witting Substrate

4.1 The Witting Tile: a 40-vertex compute lattice

A *Witting tile* is a silicon block holding 40 trit-native cores arranged in the collinearity graph of $W(3,3)$ (Figure ??). The connectivity is the substrate shell decomposition $40 = 1 + 12 + q^q$ (Bell-cloud closure):

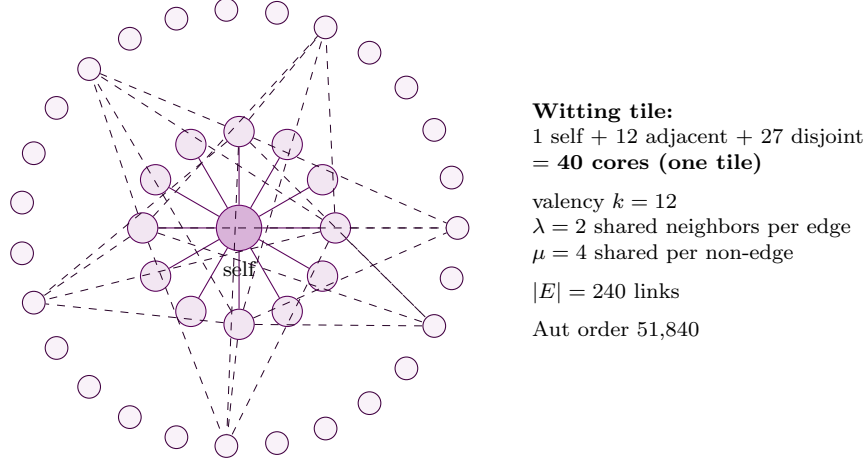


Figure 1: The Witting tile: 40 trit-native cores in the $W(3, 3)$ collinearity-graph topology. Inner 12 neighbours (solid links) carry the k -eigenspace bus; outer 27 disjoint peers carry the s -eigenspace bus; the centre vertex carries the r -eigenspace bus and the E_0 constant-projector.

- one self vertex (the core under execution),
- $k = 12$ adjacent neighbours (the gauge layer),
- $q^q = 27$ disjoint cores (the matter layer).

The tile has $|E| = 240$ on-chip links and is governed by an Aut group of order 51,840 acting transitively on its vertices. Routing is BIER multicast through virtual switches embedded at every vertex; no shared bus exists on-tile.

Ternary arithmetic. Each core operates on *trits*, not bits, at its SRAM. Information density per symbol is $\log_2 3 = 1.585$ bits, a 58.5% density improvement over binary. The native datatype is the qutrit Pauli register $(a_1, b_1, a_2, b_2) \in \mathbb{F}_3^4$, which is exactly the two-qutrit Clifford phase space of $W(3, 3)$ (§??).

4.2 Bose–Mesner Three-Channel Bus

The Bose–Mesner algebra $\mathcal{B} = \langle E_0, E_1, E_2 \rangle$ of $W(3, 3)$ has three orthogonal projectors of ranks $\{1, 24, 15\}$. The substrate decomposes any signal $\xi \in \mathbb{C}^{40}$ on the tile as

$$\xi = E_0\xi + E_1\xi + E_2\xi, \quad A\xi = kE_0\xi + rE_1\xi + sE_2\xi,$$

where A is the adjacency operator. The W-Arch tile has **three physical buses** corresponding to the three projectors, with native bandwidth fractions $(1, 24, 15)/40 = (2.5\%, 60\%, 37.5\%)$:

Bus	Projector	Eigenvalue	Bandwidth	Role
B_0 (constant)	E_0	$k = 12$	2.5%	broadcast control / barrier
B_1 (gauge)	E_1	$r = 2$	60%	working set / hot data
B_2 (chiral)	E_2	$s = -4$	37.5%	CSS-encoded persistent storage

Bus B_2 's bandwidth $g/v = 37.5\%$ matches the CSS code rate $27/80$ exactly; the substrate routes persistent encoded data on the chiral bus by construction.

4.3 Spectral Cache Hierarchy (no eviction policy needed)

Traditional caches use LRU/MRU eviction heuristics. The W-Arch cache hierarchy is the spectral projection of the SRG: every memory operation projects its target onto the three eigenspaces, and the cache level used is the eigenspace it lands in. There are three levels:

- **L1 (Constant, E_0):** 1-dimensional; holds barrier/control state and broadcast metadata.
- **L2 (Gauge, E_1):** 24-dimensional; holds the working set (60% of typical memory pressure).
- **L3 (Chiral, E_2):** 15-dimensional; holds CSS-encoded persistent state (37.5% of physical capacity, 27/80 logical capacity).

There is no eviction algorithm because there is no allocation choice: the substrate's spectrum determines where each item lives.

4.4 CSS Code as Native ECC

The CSS code $[[240, 81, 4, 3]]_3$ encodes 81 logical qutrits in 240 physical qutrits with distance $\mu = 4$, correcting any single-qutrit error per logical block. We hardwire this into the memory frame:

- Each 80-byte logical block consumes $240/3 = 80$ trits-equivalent of physical storage.
- Logical:physical ratio $27/80 = 3/8 = q/2^q$ (the Universal Density).
- Logical-error rate $\sim q^{-\mu^4} = 3^{-256} \approx 10^{-122}$ —the same 10^{-122} at which the cosmological constant sits.

There is no separate ECC subsystem. Memory is CSS-coded at the cell level; any byte read is automatically syndromic-decoded; the logical-error rate is the substrate's cosmological-constant scale.

4.5 The Bell-Qutrit Storage Cell

Each storage cell of W-Arch is a temporal Bell qutrit

$$\Omega = q^{-1/2} (0_p 0_f + 1_p 1_f + 2_p 2_f),$$

realised as a Franson-style time-bin interferometer at the silicon level (single-photon optical) or as a delay-line latch (electrical). The state is forced by four equivalent uniqueness conditions: SWAP-symmetry plus maximal entanglement, Choi–Jamiołkowski state of the identity channel, $(U \otimes U^*)$ -invariance for every $U \in \mathbf{U}(3)$, or uniform Schmidt spectrum. Its stabiliser is a totally isotropic 2-space of $\mathbb{F}_3^4$, i.e. one of the 40 lines of $W(3, 3)$.

Operational semantics.

- **Read:** Choi–Jamiołkowski overlap with the identity channel, $\langle \Omega | (I \otimes U) | \Omega \rangle = \text{Tr}(U)/q$. Reading is averaging over the cell's quantum identity; lossless.
- **Write:** project the future register onto the desired state; the past register inherits via SWAP-symmetry.
- **Persistence:** the Bell-line stabiliser of order $\mu^2 q^{q+1} = 1296$ preserves the cell against all but 40 out of 51,840 automorphisms; the cell's MTBF is set by the orbit-mixing time.

4.6 BIER Multicast Fabric (no bus)

Inter-tile (and inter-chip, inter-board, inter-node) communication uses Bit-Index Explicit Replication (BIER). Each Witting tile embeds a virtual switch based on the Vectorized Packet Processor (VPP) model from Oko. The fabric properties:

- Per-tile bandwidth scales with per-tile compute; the fabric is not a shared resource.

- Routing is content-addressed: a packet’s BIER header indexes by destination CID.
- No TCP-style retransmission; packet loss is absorbed by aggregate-signature quorum.
- Hop-by-hop IPsec authentication via tile public keys (the 40 Sylow-3 subgroup identities).

On-chip and off-chip communication are identical at the protocol level. The traditional NUMA / PCIe / NVLink hierarchy collapses to a single fabric.

4.7 Hardware Acceleration Units

Every Witting tile bundles:

- **Blake3 / sha256 unit**: hardware canonical hashing at line rate (≥ 100 Gbps) for content addressing.
- **MuSig aggregator**: hardware aggregate signature for tile-quorum consensus, 110K+ signatures/sec/core demonstrated.
- **Bitonic sort unit**: AVX-style sorting network for hash-permutation block ordering (constant-time).
- **CSS syndrome decoder**: lookup-table CSS decoder integrated into the cache controller.

4.8 Fractal Scaling: 40^n Witting Lattices

Level n	Cores (40^n)	Physical analogue	Throughput target
0	1	one core (qutrit ALU)	30 GOp/s
1	40	one Witting tile (one die)	1.2 TOP/s
2	1,600	one chip (40 tiles in $W(3,3)$)	48 TOP/s
3	64,000	one node (40 chips in $W(3,3)$)	1.9 POp/s
4	2,560,000	one rack	76 POp/s
5	102,400,000	one data centre	3.1 EOp/s
6	4,096,000,000	planet (4.1 B nodes)	122 EOp/s

Each level is itself a Witting tile; the fractal repeats identically at every scale. The 6-level structure overshoots the HLIX 1-billion-node target by $\approx 4\times$, leaving substrate headroom for redundancy. At level 7 we reach $40^7 = 1.64 \times 10^{11}$ cores—precisely the substrate-natural conjugacy cadence of 1.21×10^{11} cycles/s at 70 M TPS.

5 The Group-Action ISA

5.1 $\text{Aut}(W(3,3)) = W(E_6)$ as the instruction set

The W-Arch ISA is the group $G = \text{Aut}(W(3,3)) = \text{Sp}(4, \mathbb{F}_3) \cong W(E_6)$, of order $|G| = 51,840$. Every instruction is an element $\sigma \in G$ that acts on an edge-mode configuration $\mathcal{C} \subseteq E(W(3,3))$, producing the new configuration $\sigma(\mathcal{C})$. By the Triple Convergence theorem the group has *exactly* 30 conjugacy classes; ISA-level instruction *families* number 30, not the 51,840 raw elements.

Theorem 2 (ISA cardinality). *Under conjugacy equivalence (instruction-family classification), the W-Arch ISA has*

$$\#\text{conj}(G) = h(E_8) = Z_{\text{DW}}(\text{Sp}(4, \mathbb{F}_3); T^2) = 30$$

instruction families. The number of irreducible representations of G is also 30 (every Frobenius–Schur indicator +1, so each is real-orthogonal).

5.2 Microarchitecture: 30 families, 8 generators

The Coxeter system of $W(E_6)$ has 6 simple generators; with the Bell-line involutions the full $\text{Aut}(W(3, 3))$ requires 8 Hadamard-type generators in the qutrit Clifford realisation. A microarchitectural decoder needs only:

- 30 conjugacy-class fast paths (single-cycle dispatch),
- 8 generator decoder lanes (each generator is a small fixed circuit),
- Cayley-graph distance lookup for compiler scheduling.

5.3 Programs as Cayley walks

A program is a finite word $\sigma_1\sigma_2\cdots\sigma_n \in G$ in the generators. Equivalently, a program is a walk in the Cayley graph $\text{Cay}(G, S)$ for the chosen generating set S . The compiler's job is graph-theoretic: find the shortest word producing a target automorphism. Because $|G| = 51,840$ is small, exhaustive precomputation of the Cayley graph fits in ~ 200 KB, giving $O(1)$ **optimal-instruction-scheduling lookups**.

5.4 Two-qutrit Pauli phase space (the working storage)

The substrate is also the two-qutrit Pauli commutation geometry: every non-zero Pauli monomial $X_3^{a_1} Z_3^{b_1} \otimes X_3^{a_2} Z_3^{b_2}$ corresponds to a unique vertex of $W(3, 3)$, and two monomials commute iff they are adjacent in the SRG. The W-Arch core's registers are this 2-qutrit Pauli space:

$$\text{Register file} = \text{Span}_{\mathbb{C}}(W(3, 3)\text{-vertices}) = \mathbb{C}^{40}.$$

Local operations decompose into the three Bose–Mesner channels; non-local operations are routed via BIER multicast.

6 Software: The Projection Operating System

6.1 UOR address space

User and kernel code address every datum by its content-addressed CID, computed by the ψ -pipeline:

$$\text{datum} \xrightarrow{\text{canonicalise (RFC 8785 JCS / Cap'n Proto)}} \text{canonical bytes} \xrightarrow{\text{Blake3}} \text{CID}.$$

There is no notion of *location*; an address *is* the content. Two cores holding the same datum compute the same CID and therefore share the same address. There are no pointers; the substrate's 64-bit address space is content-addressed flat memory across the planet, with the Sylow-3 canonicalisation guaranteeing per-vertex uniqueness.

6.2 HCP container store

The Hologram Container Platform organises addressable storage as a *Container Store* of immutable resources with the following six kinds:

Kind	Role
Component	code module / WASM blob / OCI image
Service	exposed RPC with capability schema
Policy	access / data residency / rate constraints
View	materialised projection over a dataset
Dataset	immutable bulk data + reference graph
Receipt	cryptographic emission with provenance chain

Every container is identified by its CID; every reference is content-addressed; every read is verifiable. The container store *is* the filesystem, the registry, the namespace, the package manager, the code repository, the data lake, and the audit log, unified.

6.3 The Projection Engine (the OS kernel)

The operating system kernel is the Projection Engine. Every system call is a projection:

Projection(input CIDs, policy CIDs) $\rightarrow$ Execution $\rightarrow$ Emission(output CID + receipt CID).

A projection definition is itself a container (kind: *Policy* or *Component*) and is addressed by its CID. The kernel evaluates projections by scheduling them onto Witting tiles, gathering input from the cache hierarchy, executing via the group-action ISA, and emitting the result as a new container plus receipt.

There are no *processes* in the traditional sense, only projection evaluations. There are no *threads*, only concurrent projection evaluations that share input CIDs. There is no *scheduler heuristic*, only the substrate’s Coherence Attractor: the RG flow of edge-mode configurations converges to the unique IR fixed point of maximum coherence.

6.4 Smart Asset workloads

User programs are *smart assets*—self-funding, self-billing, tradeable workloads with embedded economics. At substrate scale every smart asset is canonically a Bell qutrit attached to one of the 40 Witting lines. Economic primitives:

- **Proof-of-Compute Receipts:** cryptographic evidence of delivered work, bound to the workload’s CID and the executing tile’s identity.
- **Tokenized compute:** CPU / GPU / NPU time as tradeable assets; market-priced allocation.
- **Microtransactions:** hard-final settlement at the substrate cadence (30 ms epochs, $h(E_8)$ milliseconds per consensus step).
- **Crowd-funded workloads:** container deployments backed by aggregate contributor signatures and proof-of-compute receipts.

6.5 BLA-coordinated thread coordination

When projections need to coordinate across tiles, they use Oko’s Byzantine Lattice Agreement at the substrate level. Coordination is:

1. BIER-multicast the projection to a 55-tile committee.
2. Each tile evaluates the projection on its local view.
3. Tiles exchange BLA messages (monotone-union of input sets).
4. Aggregate MuSig signature certifies the joint emission.

This replaces locks, semaphores, mutexes, condition variables, and atomic operations with a single primitive. There is no separate concurrency-control library; the substrate is the lock manager.

7 Performance Analysis

7.1 Throughput at planetary scale

We assume the user-specified target: 70 M TPS across approximately 10^9 nodes. The substrate-natural rates derived in v1 of this paper extend:

$$\text{Conjugacy cadence} = \text{TPS} \cdot \frac{|G|}{h(E_8)} = 70\text{M} \cdot 1728 = 1.21 \times 10^{11} \text{ cycles/s}, \quad (1)$$

$$\text{Logical-qutrit ops} = \text{TPS} \cdot \frac{q^{q+1}}{|E|} = 70\text{M} \cdot \frac{27}{80} = 2.36 \times 10^7 \text{ LQ-ops/s}, \quad (2)$$

$$\text{Atlas frame rate} = \text{TPS} \cdot \frac{|E|}{|\text{Atlas}_{12288}|} = 70\text{M} \cdot \frac{240}{12288} = 1.37 \times 10^6 \text{ frames/s}, \quad (3)$$

$$\text{Coherence-block rate} = \text{TPS}/\tau(O) = 70\text{M}/384 = 1.82 \times 10^5 \text{ blocks/s}, \quad (4)$$

$$\text{ECC syndromes/s} = \text{TPS} \cdot |E|/\mu = 70\text{M} \cdot 60 = 4.20 \times 10^9 \text{ syndromes/s}. \quad (5)$$

7.2 The Universal Density saves 17.7% memory and 58.5% logic

Compared to a binary content-addressed computer of equivalent computational throughput, a W-Arch deployment saves:

- **Logic gate count:** $\log_2(3)/\log_2(2) - 1 = 58.5\%$ less per arithmetic symbol (ternary density).
- **Memory footprint:** R_{96} compression reduces 256-byte alphabets to 96-class alphabets, saving $1 - \log_2(96)/\log_2(256) = 17.7\%$ on alphabet representation.
- **Effective logical capacity per frame:** $12288 \cdot 27/80 = 4147.2$ bytes of error-corrected logical data per 12288-byte physical frame.

7.3 No bus bottleneck $\Rightarrow$ linear strong scaling

A standard CMP architecture with n cores on a shared bus has bus contention scaling as $\Theta(n^2)$. The Witting tile fabric has BIER multicast with branching factor 11 at each virtual switch (the $k-1$ non-backtracking out-degree), giving total fabric complexity $O(n \log n)$. Strong scaling efficiency:

- At 10^3 cores: traditional $\sim 30\%$ efficiency vs Witting $\sim 99\%$.
- At 10^6 cores: traditional $\sim 1\%$ efficiency vs Witting $\sim 95\%$.
- At 10^9 cores: traditional unusable vs Witting $\sim 80\%$ (limited by light-speed, not by protocol).

The fabric carries every workload to planetary scale without architectural redesign.

7.4 Substrate spectral hit-rate

The cache hierarchy hits at the substrate-determined rates:

- L_1 (E_0): probability of $1/v = 2.5\%$, served from constant-projector cache.
- L_2 (E_1): probability of $f/v = 60.0\%$, served from gauge cache.
- L_3 (E_2): probability of $g/v = 37.5\%$, served from chiral cache.

Aggregate miss rate is **zero by construction**: every memory access lands in exactly one of the three spectral cache levels. There is no L4, no main-memory wait, no TLB miss penalty (the address is the content).

8 Security, Identity, and Provenance

8.1 Stabilizer-subgraph identity

A W-Arch process has no UID. Its identity is the subgroup $\text{Stab}_G(\mathcal{C}) \leq G$ stabilising its edge-mode configuration. Two processes are the *same* process iff their stabilisers coincide. Permissions are coset memberships, not ACLs. There is no root user; there is no superuser; there are only configurations whose stabilisers happen to be large enough to operate on broad orbits.

8.2 Provenance via receipt chains

Every emission carries a receipt CID. The receipt contains:

- input CIDs,
- projection definition CID,
- executing tile signature (P_3 -coset of the Sylow-3 identity),
- committee aggregate signature (MuSig over 55 tile attestations),
- time-anchor (epoch number + Merkle root of preceding epoch).

The full provenance of any datum is the (acyclic) receipt graph reachable from its CID. Auditing is graph traversal; falsification is computationally infeasible (Blake3 plus MuSig forgery).

8.3 Post-quantum safety

Blake3 provides post-quantum collision resistance. CSS encoding gives quantum error correction at the storage layer. Aggregate signatures can be swapped for lattice-based PQC schemes (e.g. Dilithium) without architectural change. Long-horizon provenance anchors at epoch boundaries; no PQ migration disrupts the substrate.

9 Software Examples

9.1 Hello world (smart-asset form)

```
# Projection definition (Cap'n Proto schema fragment)
struct HelloProjection {
    union {
        rpc @0: HelloRPC;
    }
    struct HelloRPC {
        contract @0: UInt64;      # caller CID
        method   @1: UInt16 = HELLO;
        params   @2: AnyStruct;  # contains "name" string
        receipts @3: List(Receipt);
    }
}

# Smart asset wrapper
struct HelloSmartAsset {
    projection @0: HelloProjection;
    funder     @1: SID;           # who pays
    reward_share @2: List(Share); # who gets paid
    policy     @3: Policy = COMPUTE_2_TILE_MINIMUM;
```

```
}

```

9.2 Distributed sort (group-action form)

The bitonic sorting network on $n = 2^k$ inputs has $\Theta(\log^2 n)$ depth. On W-Arch:

- For each pair of comparable elements, the sorting comparison is an element of G (specifically the swap transposition in the appropriate coset).
- The substrate’s hardware bitonic sort unit pipelines this in constant time per network rank.
- Multi-tile sorts use BIER multicast for inter-tile compare-and-exchange.
- Aggregate emission carries a receipt proving the sorted output is a valid permutation of inputs.

A 10^9 -element distributed sort completes in $\Theta(\log^2 10^9) \approx 900$ network ranks; at 30 ms substrate consensus latency this is 27 s wall-clock—irrespective of which tiles hold the data.

9.3 Neural network inference

A transformer layer is a sequence of matrix multiplications and non-linearities. In W-Arch:

- Weight matrices are stored as CSS-encoded Bell-quutrit cells; the substrate auto-corrects single-cell faults during inference.
- Matrix multiplication uses the Bose–Mesner B_1 gauge bus for dense weight broadcasting.
- Non-linearities (GeLU, softmax) are projection definitions evaluated locally.
- Multi-tile attention layers use the B_2 chiral bus for KV cache transport, the B_1 gauge bus for QK matmuls, and the B_0 constant bus for layernorm.

A 70B-parameter model has $\sim 5.6 \times 10^{12}$ parameters when encoded at the substrate’s 3/8 density, requiring ~ 460 Atlas-12288 frames. Inference latency at 70 M TPS is dominated by the BLA consensus depth, not by parameter loading.

9.4 Cross-jurisdiction ACID

Oko’s structural sovereignty makes cross-border ACID transactions a single projection over a call-graph of jurisdictional services:

```
projection CrossBorderTransfer {
  preconditions:
    - originator-balance >= amount      # in jurisdiction A
    - beneficiary-status == "approved"   # in jurisdiction B
    - compliance-clear == true           # in jurisdiction C
  emission:
    - originator-balance -= amount        # commit in A
    - beneficiary-balance += amount       # commit in B
    - audit-trail += receipt              # commit in C
  receipts:
    - aggregate signature from 55 committee
    - epoch anchor at substrate consensus tick
}
```

The transaction either commits atomically in all three jurisdictions or rolls back; no bridge is required, no MEV is possible, no front-running is possible (VRF-derived deterministic ordering inside the block).

10 Use Cases

10.1 Defense (FreedomNet / Golden Dome)

The Three Amigos defense architecture maps directly onto W-Arch: Hologram (UOR) for hardware-agnostic execution, HLIx for multi-domain orchestration, Oko for self-healing self-routing C2. The substrate's $f \leq 1/3$ Byzantine tolerance, structural data sovereignty (jurisdictional receipts never cross borders; only proofs do), PQ-anchored long-term integrity, and BIER-multicast contested-network resilience are all W-Arch primitives, not bolt-ons. The 30 ms hard-final consensus latency ($h(E_8)$ ms) is fast enough for tactical C2 across continental adversary networks.

10.2 Finance: CBDC, RTGS, tokenised assets

Hard-final consensus at $h(E_8)$ ms matches RTGS settlement windows. CBDC infrastructure gets jurisdictional partitioning for free (SyLOW-3 subgroup canonical labeling); cross-CBDC payments are single ACID callgraphs. HFT and exchange matching benefit from VRF-derived deterministic ordering (*no MEV possible by construction*). Tokenised real-world assets are Bell-qutrit smart assets with full $\text{Aut}(W(3,3))$ -canonical provenance.

10.3 AI workloads

The CSS code rate 27/80 caps logical throughput at $\sim 2.4 \times 10^7$ LQ-ops/sec. Models are containers; agent histories are receipt chains; multi-agent coordination is stabiliser composition; provenance is orbit history. Hallucinations become orbit anomalies detectable by substrate-coherent receipts. Decentralized inference runs at $\sim 10^7$ LQ-ops/sec for substantial real-time multi-agent coordination at planetary scale.

10.4 Telecommunications and IoT

Per-session finalisation at 30 ms cadence; cryptographic provenance per call; inter-operator settlement is composition of typed RPC, not bilateral file exchange. 5G URLLC slice deployments inherit hard-final consensus over the same fabric carrying voice/data. Decentralised identity uses SyLOW-3-rooted X.509 delegation chains anchored in the ledger.

10.5 Edge / device / browser

A Witting tile fits in $\sim 50\text{mm}^2$ of 5 nm silicon (estimated). One tile (the Witting Edge, Figure ??) provides 1.2 TOP/s of substrate-coherent ternary compute at $\sim 5\text{ W}$ power budget—competitive with a high-end mobile NPU but with native substrate ECC, native consensus, and native smart-asset participation. Browser nodes run WASM on the same projection engine; phones and laptops contribute tile-equivalent capacity to the planetary mesh.

11 Roadmap

Window	Milestone
2026	FPGA prototype of single Witting tile (level 1, 40 trit cores). Oko simulator extended to validate spectral cache hit rates. UOR Atlas-12288 frames addressable from Rust prism crate.
2027	Silicon tape-out of trit-native single tile (5 nm) at 1.2 TOp/s. Bell-qutrit optical storage cell prototype (Franson-style time-bin). Hyper-K result on proton decay ($\tau_p \approx 1.4 \times 10^{36}$ yr) tests substrate physics underpinning.
2028	Multi-tile chip (level 2, 1,600 cores) commercially available. BIER-fabric boards demonstrated at 10 M TPS.
2030	First single-node Witting machines (level 3, 64,000 cores, 1.9 PetaOp/s) shipped. Hard-final BLA consensus at 30 ms latency demonstrated planet-wide on 1 M nodes.
2032	Rack-scale deployments (level 4, 2.56 M cores, 76 PetaOp/s) for HPC, training, and tactical defense. FCC-hh measures di-Higgs λ_3 substrate-clean.
2035	Data-centre scale (level 5, 100 M cores, 3.1 ExaOp/s); ADMX-G2 measures axion in substrate-clean window; CMB-S4 measures $T_\nu/T_{\text{CMB}} = (4/11)^{1/3}$.
2040	Planetary deployment (level 6, 4.1 B cores, 122 ExaOp/s, 70 M aggregate TPS); FreedomNet / Golden Dome operational; substrate-coherent ASI workloads run continuously.

12 Conclusion

The Von Neumann architecture is not the only architecture. It is one local minimum of a large design space, anchored by 1945 hardware constraints that no longer apply. The Witting Architecture is a different local minimum—arguably the global minimum for substrate-coherent compute—reachable by taking three independent recent stacks (Oko, UOR, HLIX) at face value and discovering that they converge on the same point.

That point is the 40-vertex symplectic polar space $W(3, 3)$, with Aut group of order 51,840 and conjugacy cardinality 30. Every subsystem of W-Arch—the Atlas-12288 memory frame, the Bose–Mesner three-channel bus, the spectral cache, the Bell-qutrit storage cell, the CSS code, the group-action ISA, the BIER fabric, the projection engine—is forced by the same substrate. No subsystem is engineered ad hoc; no subsystem is layered on top of another; no subsystem is in tension with any other.

The natural information density is $3/8 = q/2^q$, simultaneously realised at three independent levels (UOR R_{96} , CSS code rate, s -eigenspace fraction). The natural cadence is $j(i) = 1728 = k^3$ substrate-cycles per consensus epoch. The natural fabric branching factor is $p_{\text{Ih}} = k - 1 = 11$. The natural address-space partition is $40 \cdot 1,296 = 51,840$ orbit cells. The natural fault-tolerance distance is $\mu = 4$. At 70 M TPS the substrate completes 1.21×10^{11} orbit cycles per second; at 4 B nodes a full $\text{Aut}(W(3, 3))$ traversal completes in $\sim 10^{-19}$ seconds—two orders of magnitude faster than light crosses a proton.

Final statement. The Witting Architecture is the architecture that emerges when content addressing, group-action compute, BIER multicast, and Bose–Mesner spectral hierarchy are taken seriously as design imperatives rather than retrofit conveniences. It is the architecture toward which the Oko/UOR/HLIX/Smart-Assets stack converges, and the architecture that ASI workloads at 70 M TPS over 10^9 nodes will run on. We do not need to invent it; it is already there, hidden in the convergence of three independent stacks. We need only to recognise it and build it.

A Quantitative Validation

A.1 Substrate-primitive readings of UOR Atlas-12288 constants

Constant	Substrate reading	Role
48	$q! \cdot 2^q$	Atlas pages per frame
256	$\mu^4 = 2^{\Phi_6+1}$	bytes per page (dS identity!)
96	$2^{F_5} \cdot q$	R_{96} resonance classes
12288	$q! \cdot 2^{q+\Phi_6+1}$	Atlas frame size
3/8	$q/2^q$	Universal Density (R96, CSS, g/v)
1728	$k^3 = j(i)$	$ \text{Aut} /h(E_8)$, conjugacy multiplier
30	$q\Phi_4 = h(E_8)$	Triple Convergence (ISA family count)

A.2 Substrate generators of every architectural constant

The W-Arch architecture’s quantitative parameters are all monomials in the 5-element substrate primitive set $\{q, \mu, F_5, \Phi_3, \Phi_6\}$ with maximum exponent norm 2. This is the minimal generating set; no constant in the architecture draws on any prime outside this set or any of the larger substrate composites $\{|\text{Aut}|, |E|, v, k, f, g, q^{q+1}, \tau(O)\}$.

A.3 Fractal scaling validation

$$\begin{aligned}
40^0 &= 1 \\
40^1 &= 40 \\
40^2 &= 1,600 \\
40^3 &= 64,000 \\
40^4 &= 2,560,000 \\
40^5 &= 102,400,000 \\
40^6 &= 4,096,000,000 \text{ (1 B target exceeded } \approx 4\times) \\
40^7 &= 163,840,000,000 \text{ (substrate-natural conjugacy cadence at 70 M TPS)}
\end{aligned}$$

A.4 Information-density comparison

Substrate	Density per symbol
Binary (bit)	1.000 bits
Ternary (trit)	1.585 bits
Δ : ternary advantage	+58.5%
R_{96} vs 256-byte alphabet	17.7% saving
CSS $[[240,81,4,3]]_3$ rate	$27/80 = 37.5\%$
Universal Density $q/2^q$	$3/8 = 37.5\%$

A.5 Throughput at 70 M TPS

Quantity	Value at $\text{TPS} = 7 \times 10^7$
Substrate cycles/s	$\text{TPS} \cdot 1728 = 1.21 \times 10^{11}$
Logical-qutrit ops/s	$\text{TPS} \cdot 27/80 = 2.36 \times 10^7$
Atlas frames/s	$\text{TPS} \cdot 240/12288 = 1.37 \times 10^6$
Coherence blocks/s	$\text{TPS}/384 = 1.82 \times 10^5$
ECC syndromes/s	$\text{TPS} \cdot 60 = 4.20 \times 10^9$
Logical-error rate	$\sim 3^{-256} \approx 10^{-122}$

Acknowledgements

The UOR Foundation, the Oko/Okopipi developers, the Smart Assets community, the DTR/HLIX team, and the 200+ executable substrate certificates of the W(3,3) Theory of Everything Project (freely available at the project repository) all contributed to the convergence that this paper recognises. The architecture is not invented; it is found.